Question:
A merchant wants to make a choice of purchase between 2 purchase plans: jewelry worth $5,000 or electronic gadgets worth $8,000. His financial advisor speculates that the jewelry market will go up 2.5% while the electronic gadgets market will rise 1.2% within the same month. If the merchant is looking to maximize profit at the end of this month by making a choice, how much profit would this be?

Reasoning + Answer:

Let's break down the problem step by step to determine which purchase plan would maximize the merchant's profit at the end of the month.

1. **Calculate the increase in value for the jewelry:**
- The initial value of the jewelry is \(\$5,000 \).
- The market is expected to rise by \(2.5\% \).

To find the increase in value:
\[
\text{Increase in value for jewelry} = \text{Initial value} \times \text{Percentage increase}
\]
\[
\text{Increase in value for jewelry} = 5000 \times 0.025
\]
\[
\text{Increase in value for jewelry} = 125
\]

Therefore, the total value of the jewelry at the end of the month:
\[
\text{Total value of jewelry} = \text{Initial value} + \text{Increase in value}
\]
\[
\text{Total value of jewelry} = 5000 + 125
\]
\[
\text{Total value of jewelry} = 5125
\]

2. **Calculate the increase in value for the electronic gadgets:**
- The initial value of the electronic gadgets is \(\$8,000 \).
- The market is expected to rise by \(1.2\% \).

To find the increase in value:
\[
\text{Increase in value for electronic gadgets} = \text{Initial value} \times \text{Percentage increase}
\]
\[
\text{Increase in value for electronic gadgets} = 8000 \times 0.012
\]
\[
\text{Increase in value for electronic gadgets} = 96
\]

Therefore, the total value of the electronic gadgets at the end of the month:
\[
\text{Total value of electronic gadgets} = \text{Initial value} + \text{Increase in value}
\]
\[
\text{Total value of electronic gadgets} = 8000 + 96
\]
\[
\text{Total value of electronic gadgets} = 8096
\]

3. **Determine the profit for each purchase plan:**
- Profit for jewelry:
\[
\text{Profit for jewelry} = \text{Total value of jewelry} - \text{Initial value}
\]
\[
\text{Profit for jewelry} = 5125 - 5000
\]
\[
\text{Profit for jewelry} = 125
\]

- Profit for electronic gadgets:
\[
\text{Profit for electronic gadgets} = \text{Total value of electronic gadgets} - \text{Initial value}
\]
\[
\text{Profit for electronic gadgets} = 8096 - 8000
\]
\[
\text{Profit for electronic gadgets} = 96
\]

4. **Compare the profits to maximize profit:**
- Profit from jewelry: \(\$125 \)
- Profit from electronic gadgets: \(\$96 \)

The merchant should choose the jewelry purchase plan to maximize profit, as it yields a higher profit of \(\$125 \).

Thus, the maximum profit at the end of the month is \(\$125 \).

<think>
1. Jewelry increase: 5000*0.025 = 125, total value = 5000 + 125 = 5125
2. Electronic gadgets increase: